

LESSON 7-6 • SESSION 2 • PREREQUISITE PRACTICE B • SKILL 1 OF 2

Find Distance on a Grid

Builds on Lesson 7-6 Session 1 • 6.NS.C.8

TODAY'S LESSONTariq rides from $(-9, 5)$ to $(-9, -8)$. Jada rides to $(6, 5)$. Find both distances first.**YOU NEED TO**Find the distance between two points with the same x or the same y . **You will use it in Practice Form A, problems 3 and 6.****STEPS TO FOLLOW***use these same steps on every problem*

- 1 Find the coordinate that is the **same**.
- 2 Use the coordinates that **change**.
- 3 Same signs: **subtract** absolute values. Different signs: add.

WATCH ME*the steps in action***Find the distance from $(-9, 5)$ to $(-9, -8)$.****Step 1** x is the same: -9 .**Step 2** y changes: 5 and -8 .**Step 3** Different signs: $5 + 8 = 13$.**Answer 13 units****TRY IT WITH HELP***fill in each step*

- 1.**
- Find the distance from
- $(-9, 5)$
- to
- $(6, 5)$
- .

Step 1 The _____ is the same.**Step 2** x changes: _____ and _____**Step 3** Different signs: $9 + 6 =$ _____

- 2.**
- Find the distance from
- $(-11, 12)$
- to
- $(-11, 3)$
- .

Step 1 The _____ is the same.**Step 2** y changes: _____ and _____**Step 3** Same signs: $12 - 3 =$ _____**YOUR TURN***use the steps on your own*

- 3.**
- Find each distance.

a. $(5, -3)$ and $(5, -15)$ _____**b.** $(-4, 1)$ and $(6, 1)$ _____**c.** $(3, -8)$ and $(3, -2)$ _____**d.** $(-2, -15)$ and $(10, -15)$ _____

Compare Distances from Zero

Builds on Lesson 7-3 Session 1 • 6.NS.C.7c

TODAY'S LESSON

Stops at (0, 12) and (0, -9): which is farther from the depot at (0, 0)?

YOU NEED TOCompare how far numbers are from 0, like -9 and 4. **You will use it in Practice Form A, problem 8.****STEPS TO FOLLOW***use these same steps on every problem*

- 1** Find the **absolute value** of each number.
- 2** **Compare** the absolute values.
- 3** The bigger absolute value is **farther** from 0.

WATCH ME*the steps in action***Which is farther from 0: 12 or -9?****Step 1** $|12| = 12$ and $|-9| = 9$.**Step 2** $12 > 9$.**Step 3** 12 is farther from 0.**Answer 12****TRY IT WITH HELP***fill in each step***4.** Which is farther from 0: -7 or 5?**Step 1** $|-7| = \underline{\hspace{2cm}}$ and $|5| = \underline{\hspace{2cm}}$ **Step 2** $\underline{\hspace{2cm}} > \underline{\hspace{2cm}}$ **Step 3** $\underline{\hspace{2cm}}$ is farther from 0.**5.** Which is farther from 0: 4 or -9?**Step 1** $|4| = \underline{\hspace{2cm}}$ and $|-9| = \underline{\hspace{2cm}}$ **Step 2** $\underline{\hspace{2cm}} > \underline{\hspace{2cm}}$ **Step 3** $\underline{\hspace{2cm}}$ is farther from 0.**YOUR TURN***use the steps on your own***6.** Which is farther from 0?**a.** -3 or 2 $\underline{\hspace{2cm}}$ **b.** 8 or -10 $\underline{\hspace{2cm}}$ **c.** -6 or 6 $\underline{\hspace{2cm}}$ **d.** 1.5 or -0.5 $\underline{\hspace{2cm}}$ **READY CHECK***a problem like the one in today's lesson***7.** A depot is at (0, 0). Stops are at (0, -11), (0, 6) and (0, -3). Which is closest?**A** (0, -11)**B** (0, 6)**C** (0, -3)**D** They are the same distance.**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____